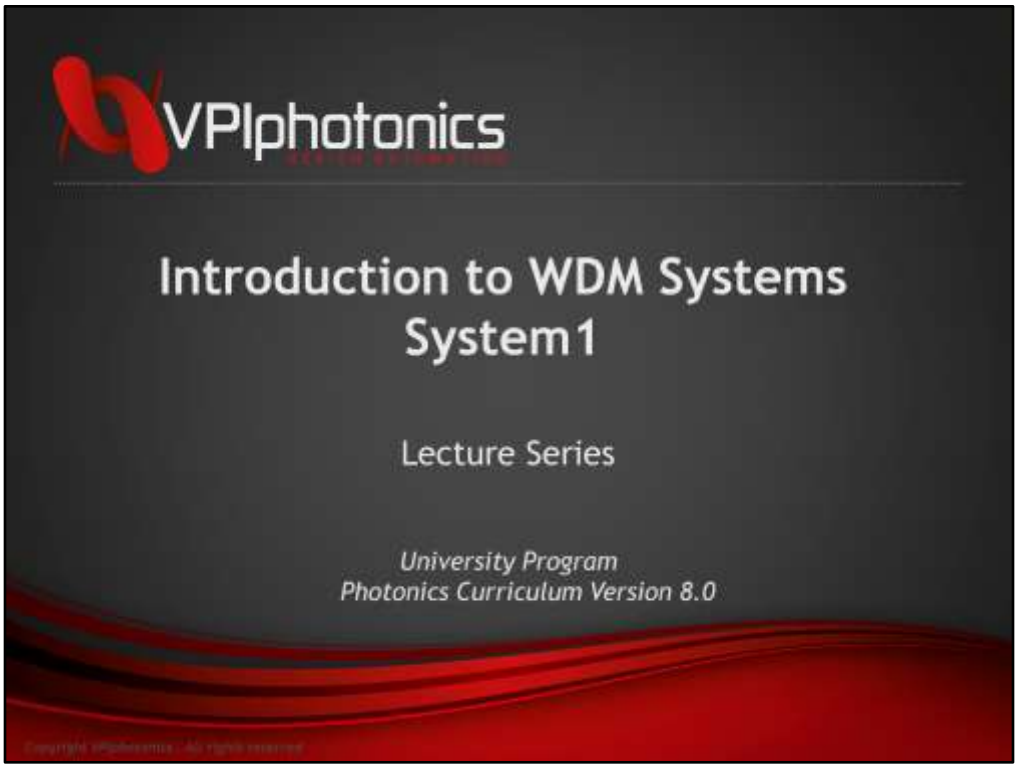
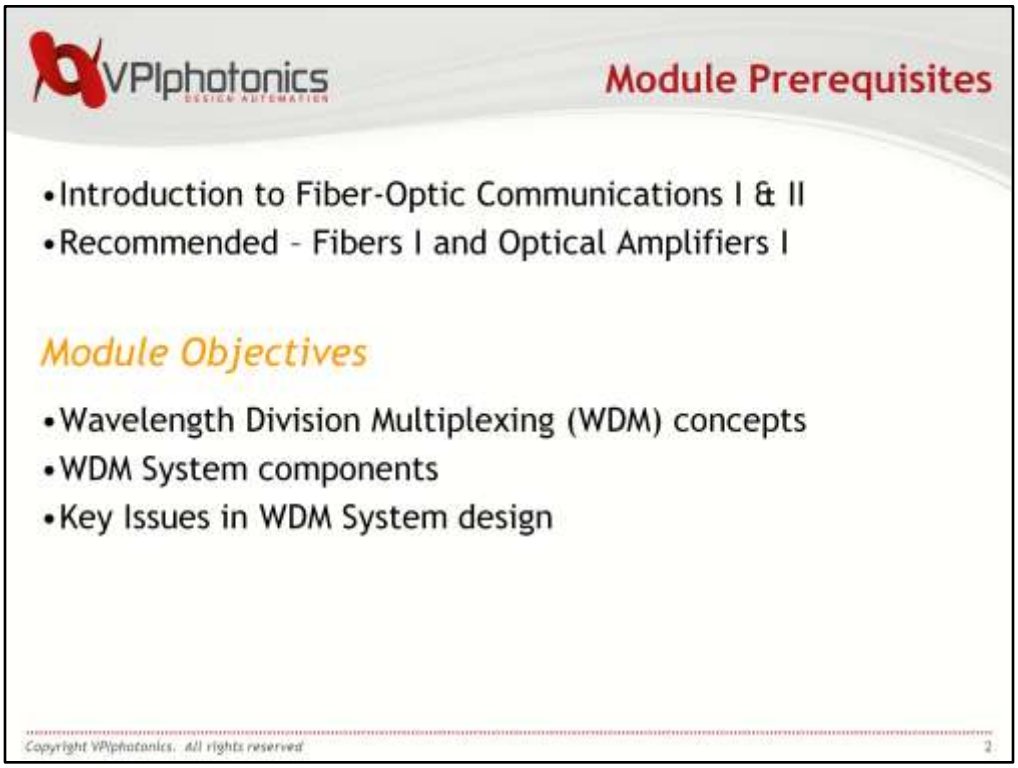
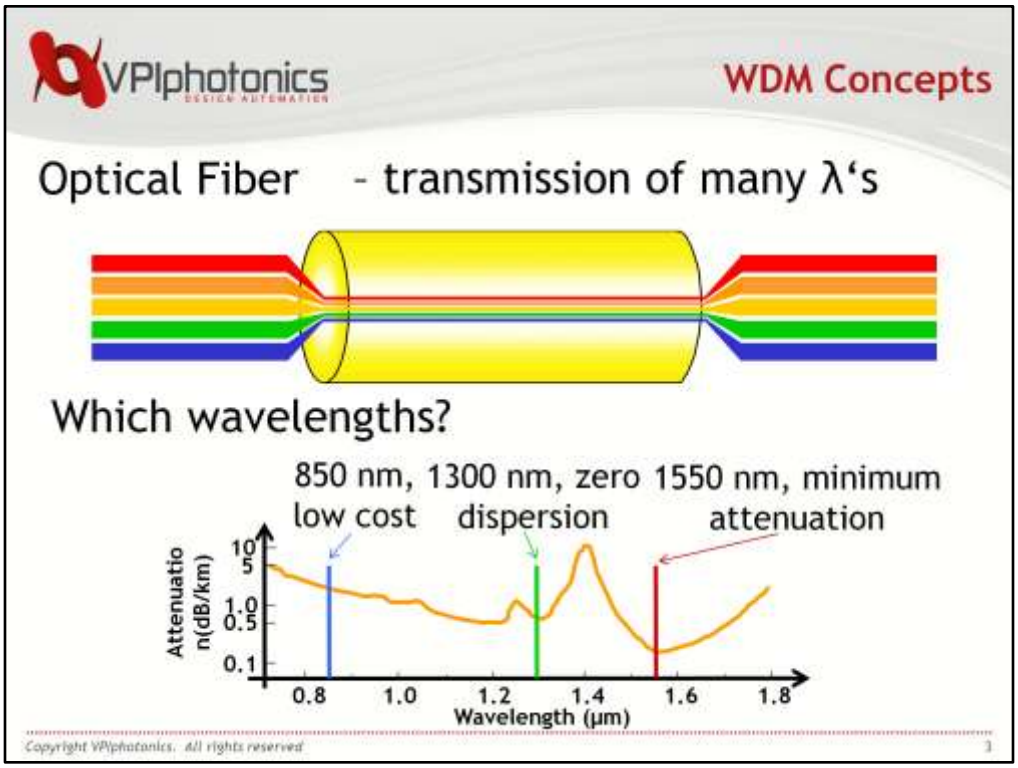

VPIphotonics
Introduction to WDM Systems
System1
Lecture Series
University Program
Photonics Curriculum Version 8.0



Module prerequisites and objectives – edit to suit the specific requirements of your course.



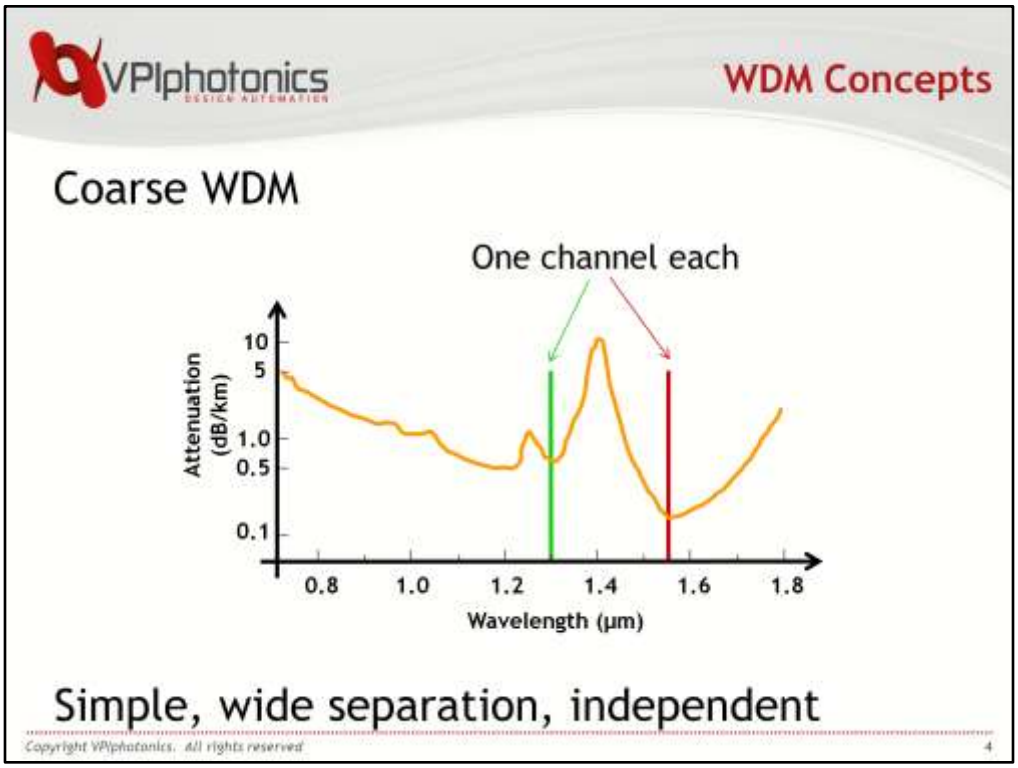
In the previous two introductory lectures, the fiber-optic systems that were discussed transmitted information via only a single wavelength channel. The information in that channel is modulated onto laser light of a particular wavelength (nominally assumed to be 1550 nm).

Optical fibers are capable of transmitting, simultaneously, information encoded in many channels, each of a different wavelength. This is very attractive, as the capacity of a fiber optic system that transmits information using multiple channels of different wavelengths has many times the capacity of a system that only uses a single channel.

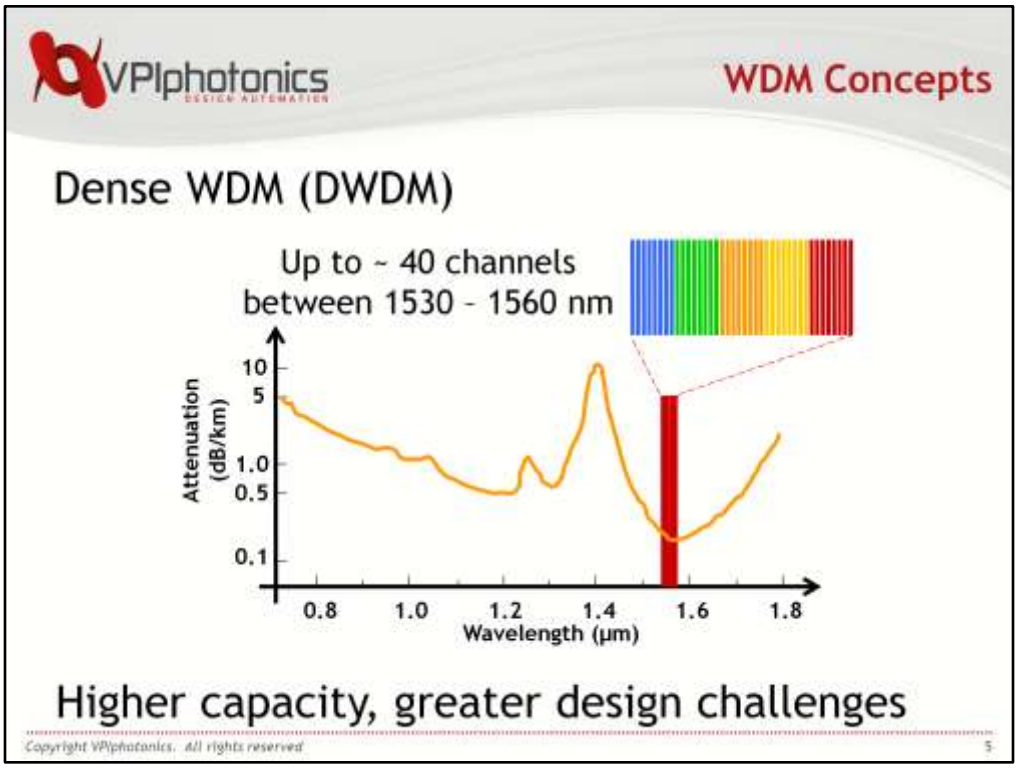
The concept of sending information using different wavelength channels is know as "Wavelength Division Multiplexing" (WDM). Students who remember their communications theory lecture will see that it is essentially the same as Frequency Division Multiplexing (FDM). In photonics, wavelength is generally the preferred unit, instead of frequency, for describing optical spectra – hence WDM is used to indicate multiplexing in the frequency domain for optical/photonic systems, whereas FDM is used for electrical systems.

Which wavelengths are suitable? This depends mainly on the attenuation and dispersion of the optical fiber used at the particular wavelength in which the data is to be transmitted, and of course, the cost involved. For the bulk of the standard optical fiber that has been deployed, there are three wavelength regions that are suitable – the attenuation windows around 850 nm, 1300 nm and 1550 nm.

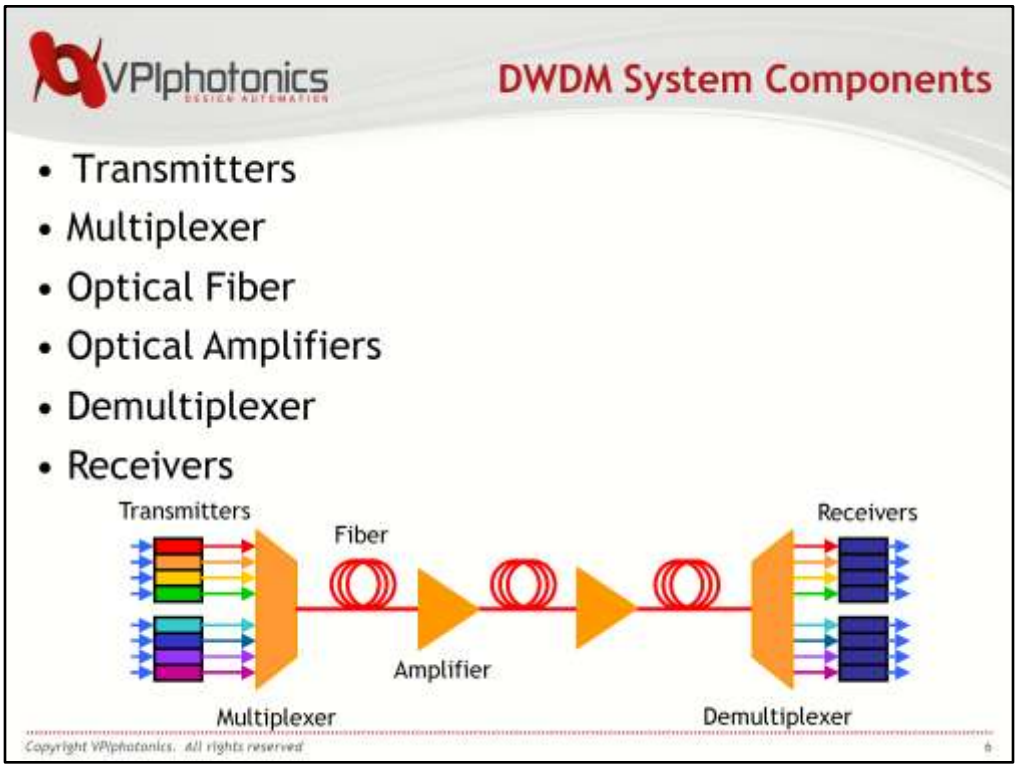
850 nm window: The technology is mature – infrared LEDs, plastic optical fibers and plastic connectors/splices are cheap, robust, reliable and easy to make. Due to the high attenuation, high dispersion and spectral width of LED light, this technology is suited for



WDM systems are also classified according to how closely spaced the different wavelength channels are. In the early days of WDM transmission, there were two wavelength channels, each at the 1300 and 1550 nm windows (850 nm is not used due to the limited transmission distance). Such a method of WDM is now usually referred to as “Coarse WDM”.

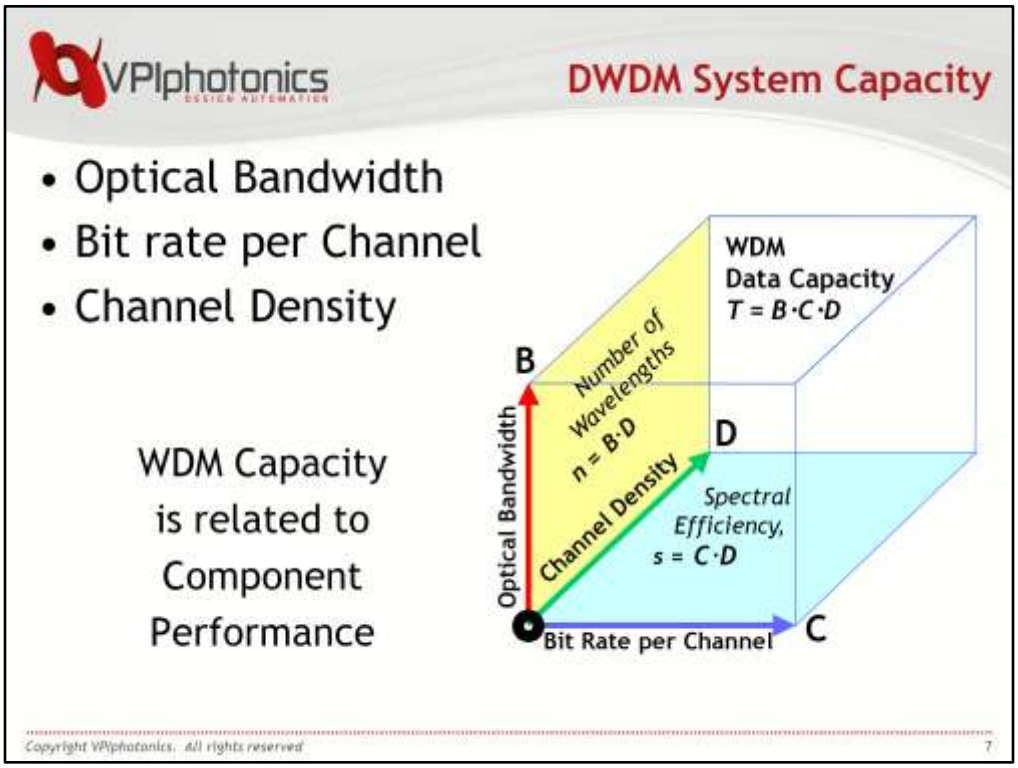


Dense WDM are WDM systems where their wavelength channels are closely spaced together. Most WDM systems of interest today are of this type. Generally, the channels are bunched together to fit into the bandwidth of an optical amplifier (for obvious reasons).



The rest of this lecture will focus on DWDM systems.

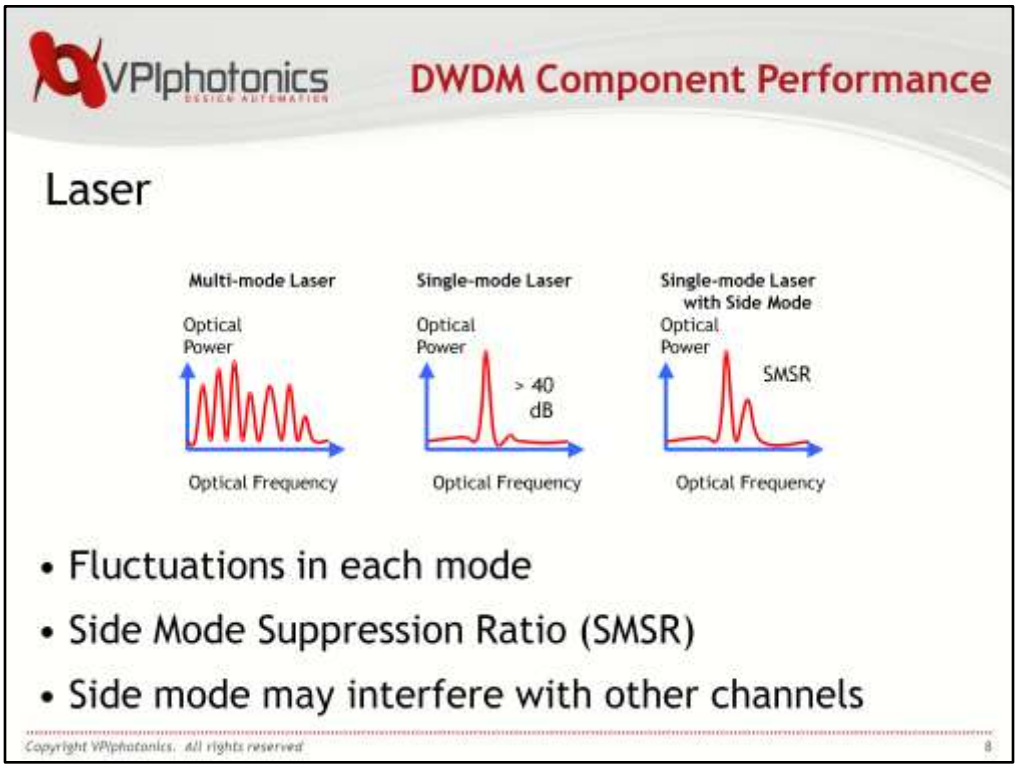
Here is a list of the components for a basic DWDM system. Apart from components such as lasers, receivers, fibers and amplifiers (which the student has encountered in the previous two modules), there are multiplexers and demultiplexers. Naturally, there are more lasers and receivers in a DWDM system compared to a single channel system.



What determines the capacity of a DWDM system?

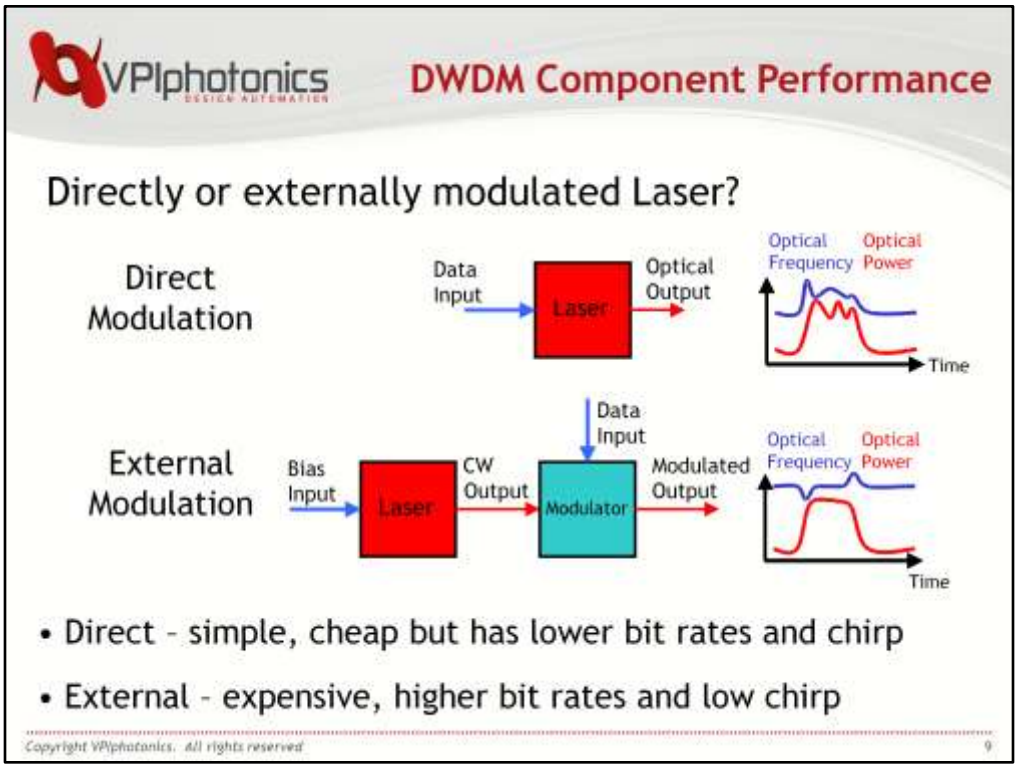
There are three key points: The optical bandwidth of the system (B), the bit-rate of each individual channel (C), and the number of channels that can be fitted within the optical bandwidth of the system, i.e., the channel density (D).

Those are conveniently represented by the three axes of a three dimensional coordinate system. The volume of the cuboid represents the total data capacity of the WDM system. The length of the sides of the cuboid is determined by the performance or specifications of the components of the DWDM system.



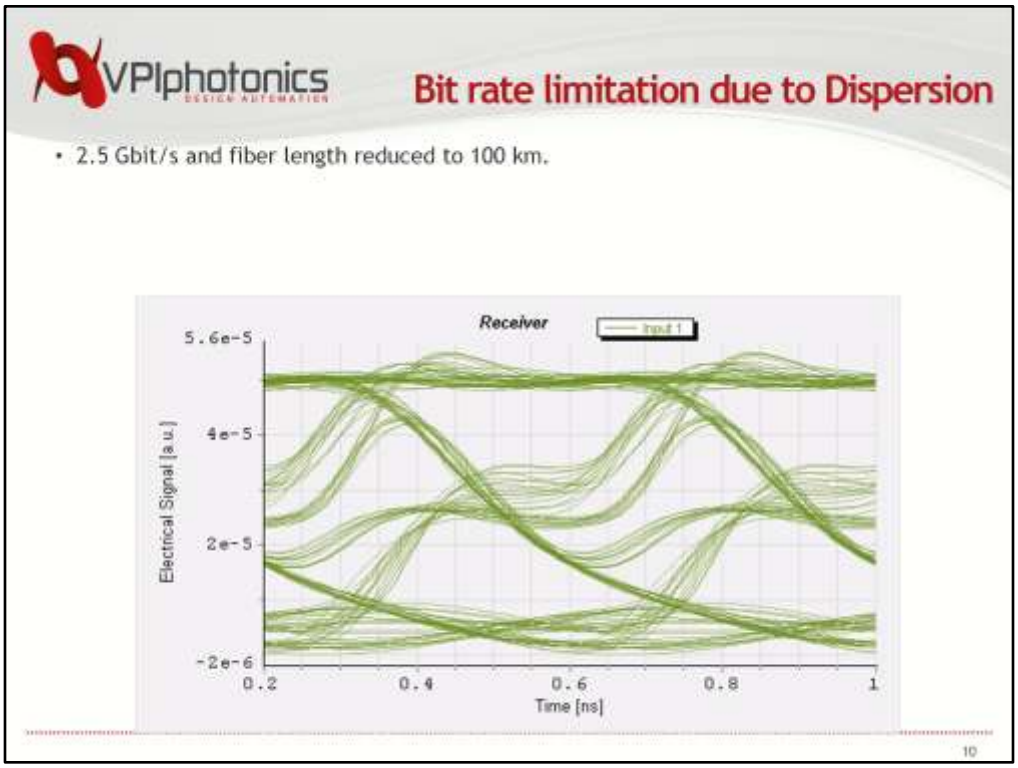
Laser

For WDM applications, single mode lasers with very high side mode suppression ratio are desirable.
Multimode lasers are not suitable, as they cannot be spaced too closely together.
A laser with low side mode suppression ratio will limit the system, due to interference from the side mode.

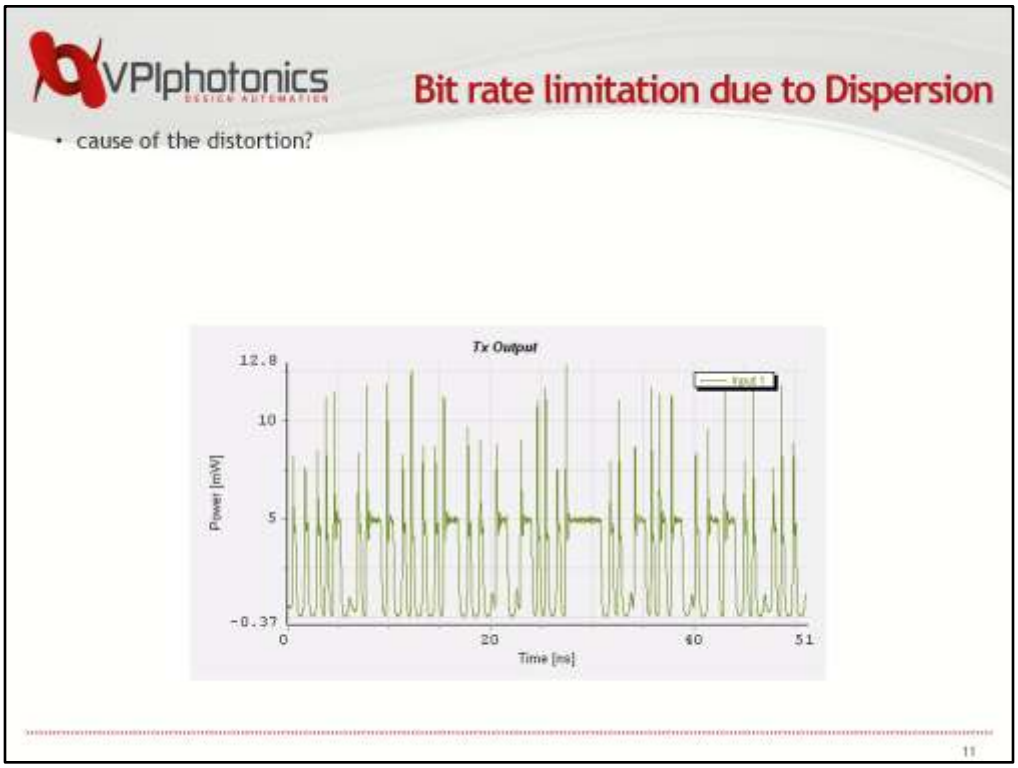


There is a trade-off here:

Directly modulated lasers are cheap, but they exhibit significant frequency chirp, which limits the transmission bit rates and also the maximum transmission distance.
Externally modulated lasers, which produce optical signals with nearly zero chirp, are more expensive.



Next, open FOC2_5. This is the same system as before, but now the bit rate is increased to 2.5 Gbit/s. The resulting eye diagram is shown below. The received pulses are distorted, leading to a closing of the eye and the BER of the system has increased above the acceptable level of $1e^{-9}$.



To find the cause of the distortion, we first want to look at the output of our transmitter to make sure that the pulses are generated properly. For this, affix a *SignalAnalyzer* to the output of the laser module:

Wire the visualizer to the transmitter (output of the laser module *LaserRateEqSM*).

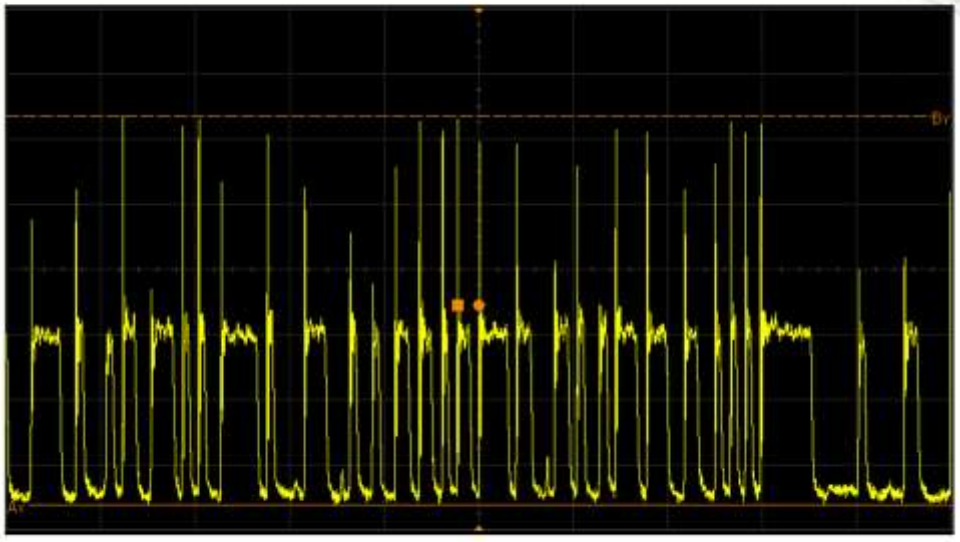
Labels of the visualizer outputs

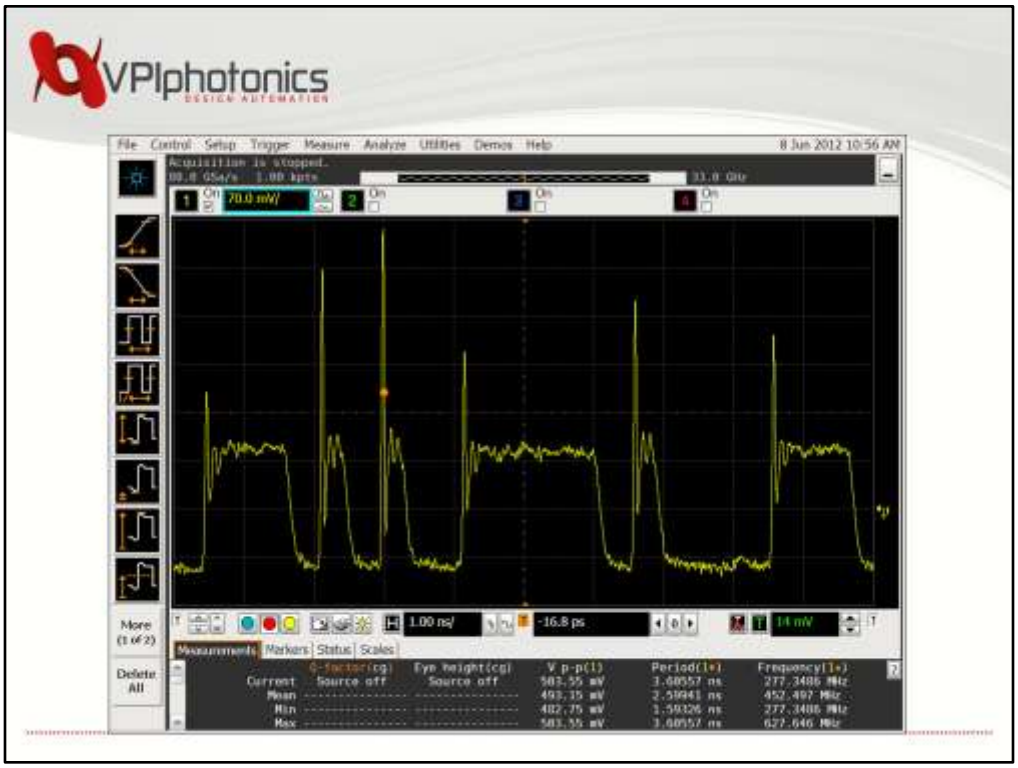
SignalAnalyzer (affixed to the transmitter)—**Title** = `Tx Output`

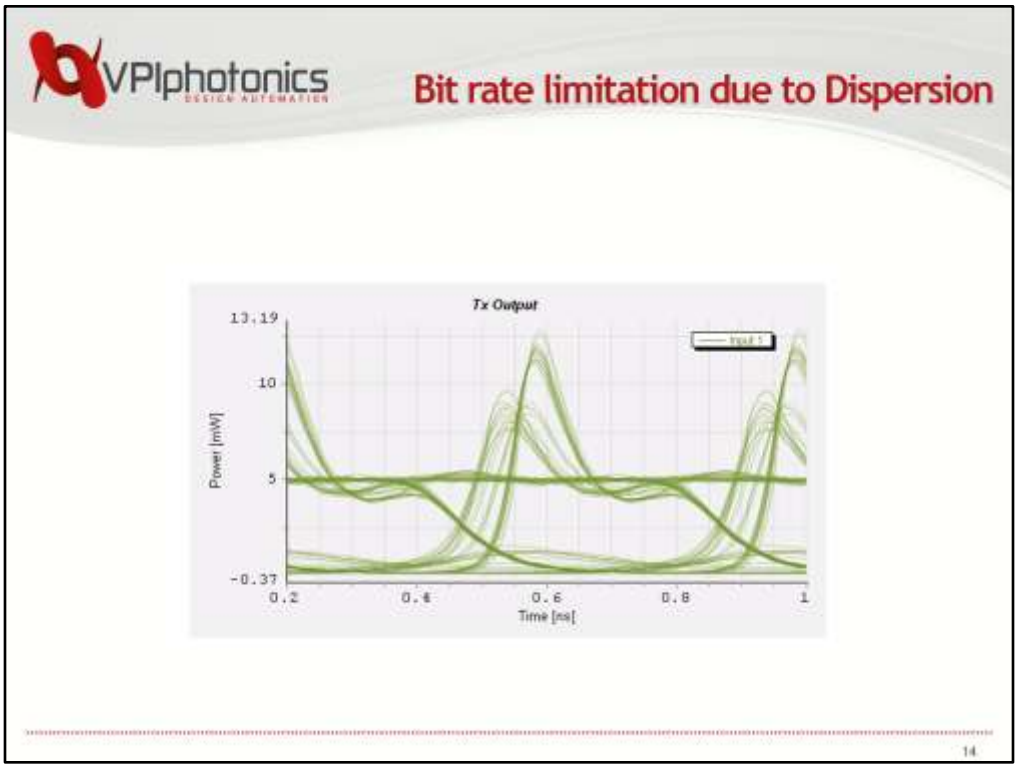
SignalAnalyzer (affixed to the receiver) —**Title** = `Receiver`.

Save your schematic and repeat the simulation.

2.5 Gbps, Tektronix,

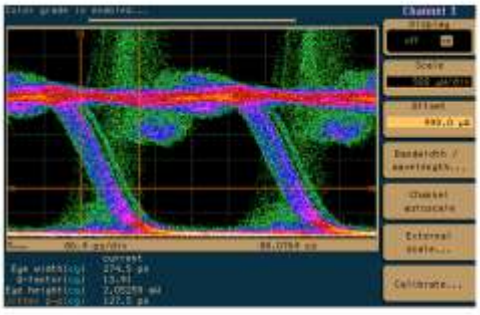


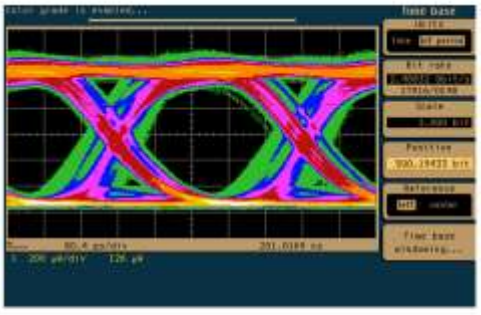

VPIphotonics
DESIGN AUTOMATION
File Control Setup Trigger Measure Analyze Utilities Demos Help
8 Jun 2012 10:56 AM
Acquisition is stopped.
80.0 GSa/s 1.00 kpts
33.0 GHz
70.0 mV/
1.00 ns/
-16.8 ps
14 mV
More (1 of 2)
Delete All
Measurements
Markers
Status
Scales
Q-Factor(cg)
Eye Height(cg)
V p-p(1)
Period(1*)
Frequency(1*)
Current
Source off
Source off
583.55 mV
3.68557 ns
277.3486 MHz
Mean
493.35 mV
2.59841 ns
452.497 MHz
Min
402.75 mV
1.59326 ns
277.3486 MHz
Max
583.55 mV
3.68557 ns
627.646 MHz

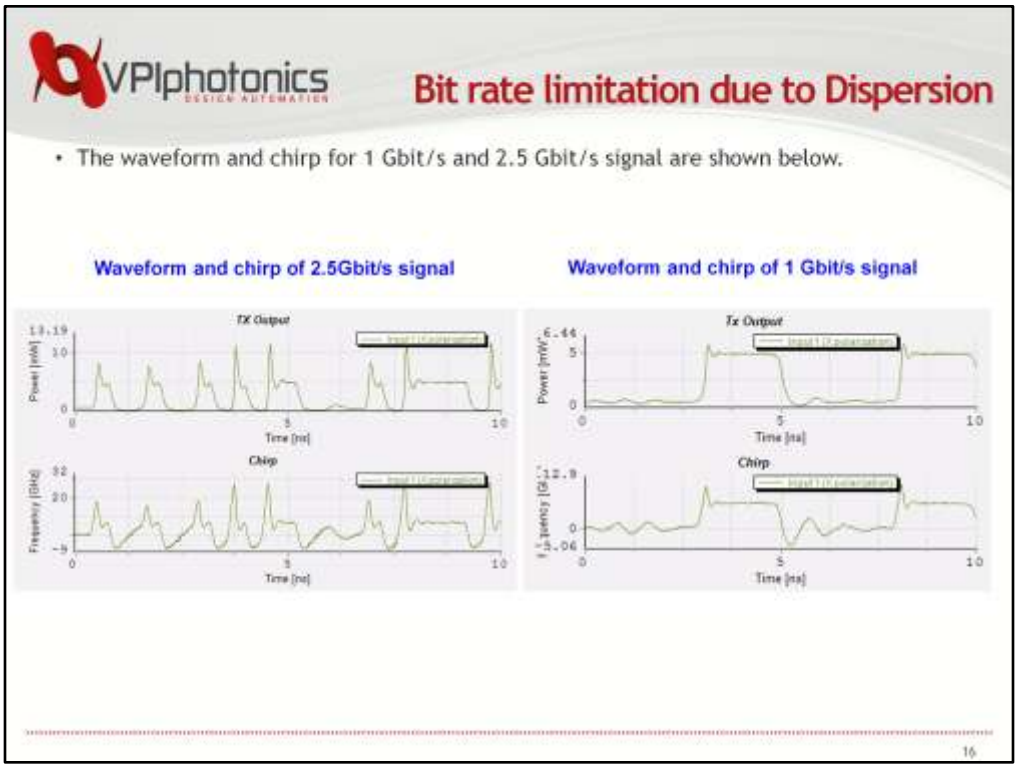


Press the eye-button in the toolbar of the VPIphotonicsAnalyzer window for the *SignalAnalyzer* corresponding to the output of the transmitter.

Evaluating the eye-diagram shows that the optical pulses still produce a fairly wide open eye, despite the overshoots. Thus, the observed degradation of the pulses at the receiver must be caused by something else.







Investigating the signal chirp (FOC2_5 and FOC2_4)

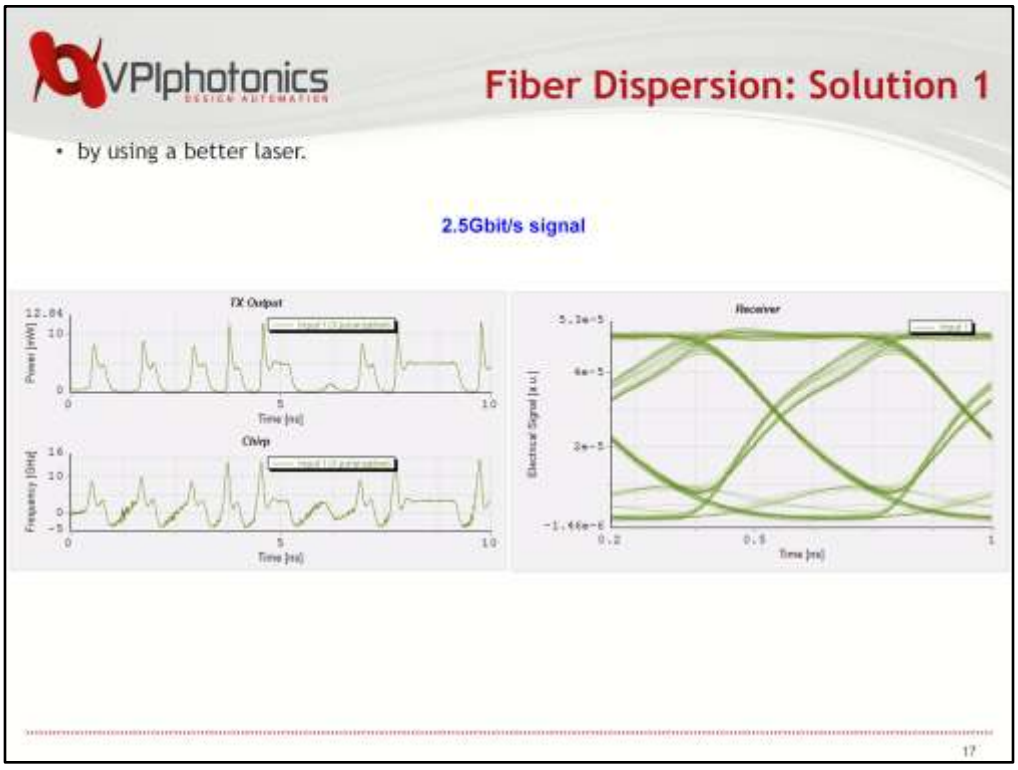
It is well known that amplitude modulation in semiconductor lasers is accompanied by phase modulation. The mode frequency differs from its steady-state value. Pulses experiencing such a phase modulation are called *chirped*.

Display the signal chirp at the transmitter output (in the *ViScope,* once the graph is displayed, select the *X* polarization in the **Power** tabpane. Then, highlight the **Chirp** checkbox).

Zoom in.

You can see from the lower trace in the Scope that the pulses are considerably chirped. That frequency chirp is particularly large at the leading edge of the pulses, exceeding 20 GHz in this example.

The frequency chirp broadens the spectrum of the optical signal considerably. Due to fiber dispersion, different frequencies propagate with different velocities, which leads to pulse distortion. With increased signal bit rates, the tolerable dispersion is decreasing as the bit durations become shorter. Thus, fiber dispersion becomes important much earlier for a bit rate of 2.5 Gbit/s than for a bit rate of 1 Gbit/s.



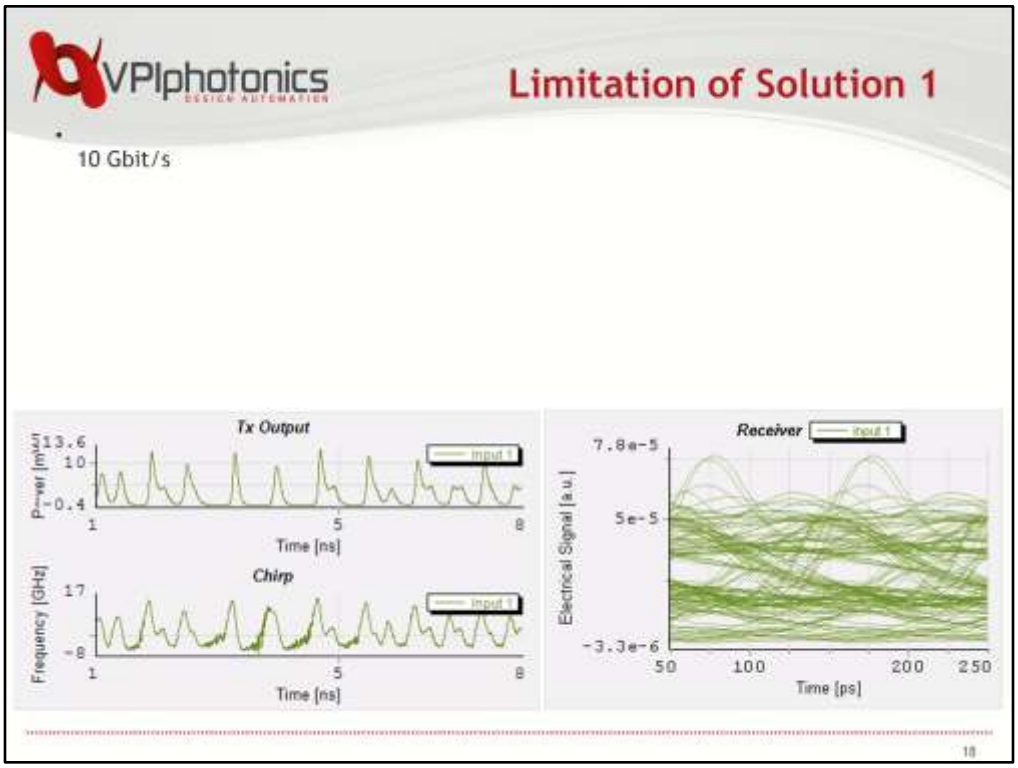
FOC2_3

The simulation results show that:

the received pulses overlap to a clear eye-diagram

the frequency chirp of the transmitter pulses is significantly reduced

the frequency chirp was indeed the limiting factor in the previous system.



Now, let's see how this improved laser works when driven at 10 Gbit/s (or use FOC2_6)

In FOC2_5, edit the following global parameters of your schematic:

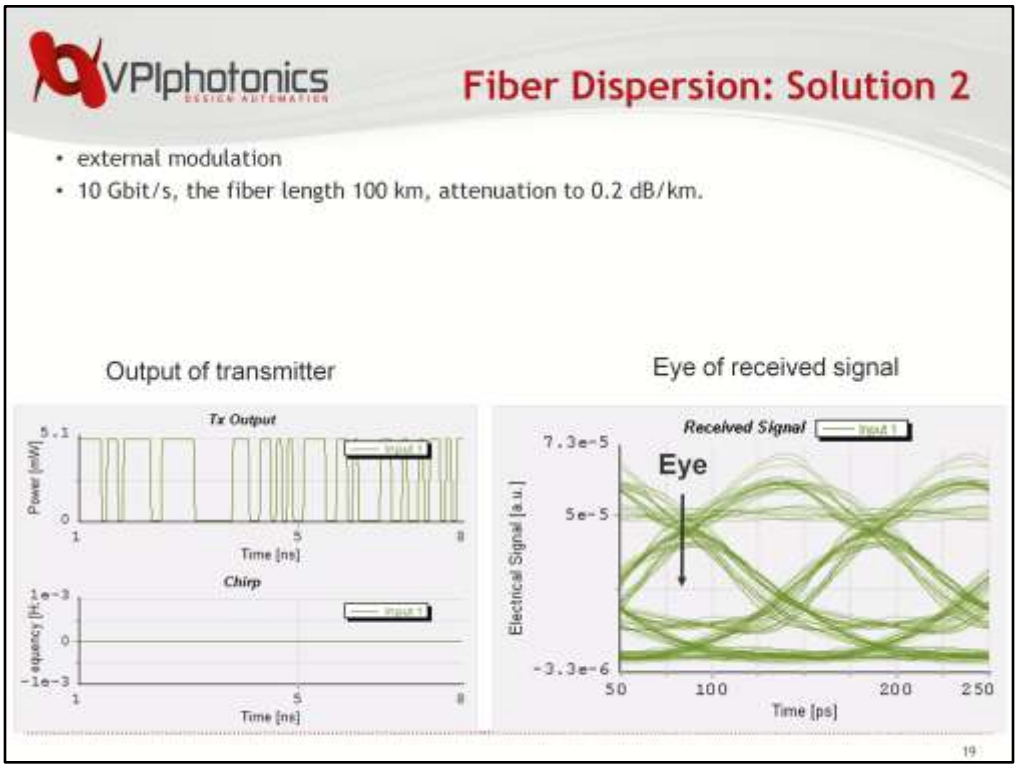
Lab2_1—**TimeWindow** = `128/10e9`

Lab2_1—**BitRateDefault** = `10e9`

Lab2_1—**SampleRateDefault** = `320e9`

`Repeat the simulation`

`The Scope` at the output of the transmitter shows already strong bit pattern effects. Obviously, the laser under consideration is not able to follow the modulation at 10 Gbit/s. The eye is already totally closed at the transmitter end.



FOC2_5

Editing parameters

Edit the following parameters of the laser transmitter to simulate an average CW power of 5 mW:

TxExtModLaser—**Laser_AveragePower** = `5e-3`

TxExtModLaser—**ChannelLabel** = `Ch1`

Also, a channel label is issued to the transmitter allowing the extraction of auxiliary transmitter information at the receiver. This is necessary to perform BER estimation later on in this lab exercise.

Save your setup and run the simulation.

Evaluating the signal shapes

It is found that the externally modulated transmitter generates clear and chirp-free pulses.

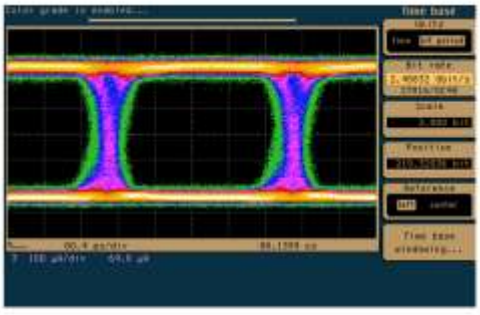
Increase the length of the SMF fiber to 120 km and repeat the simulation.

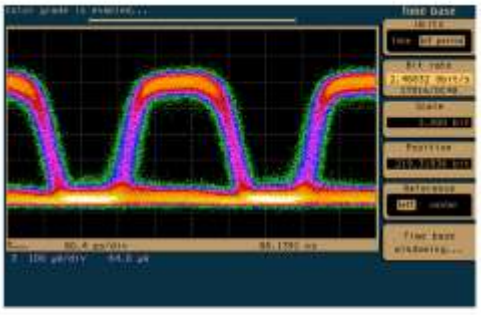
The pulses at the receiver end are distorted, resulting in large eye closure (see Hypertextigure). The observed impairments of the system performance are caused

VPIphotonics
DESIGN AUTOMATION

NRZ

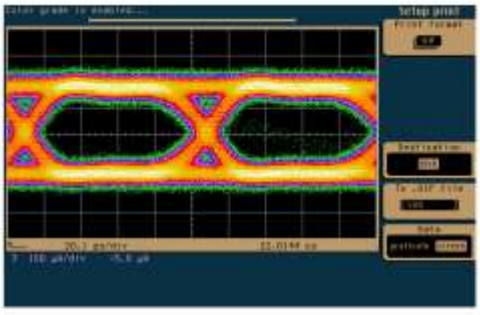
2.5 Gbps, RZ

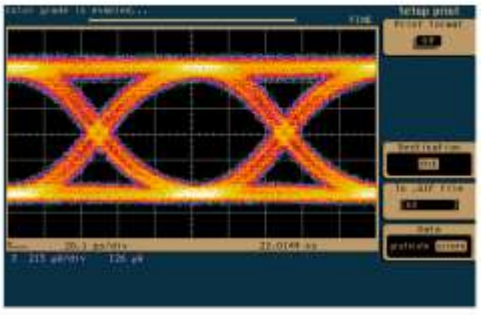




VPIphotonics
DESIGN AUTOMATION

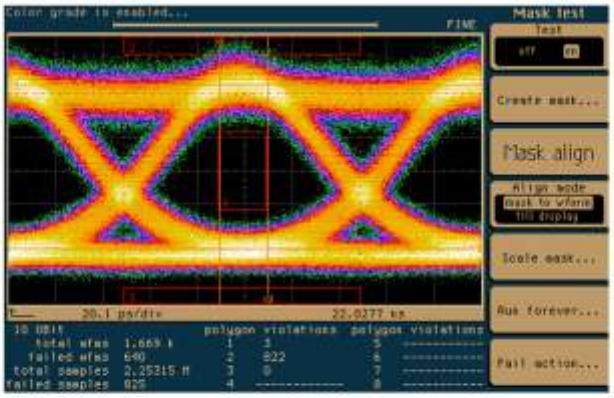
10 Gbps

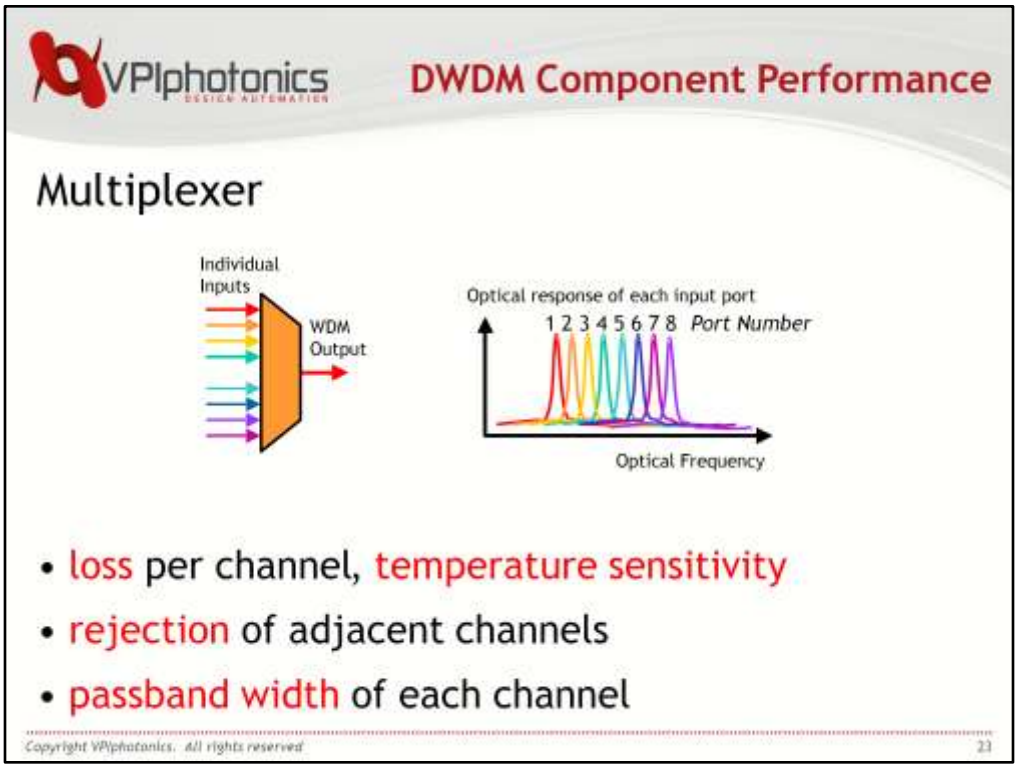




VPIphotonics
DESIGN AUTOMATION

10 Gbps, 1300 km

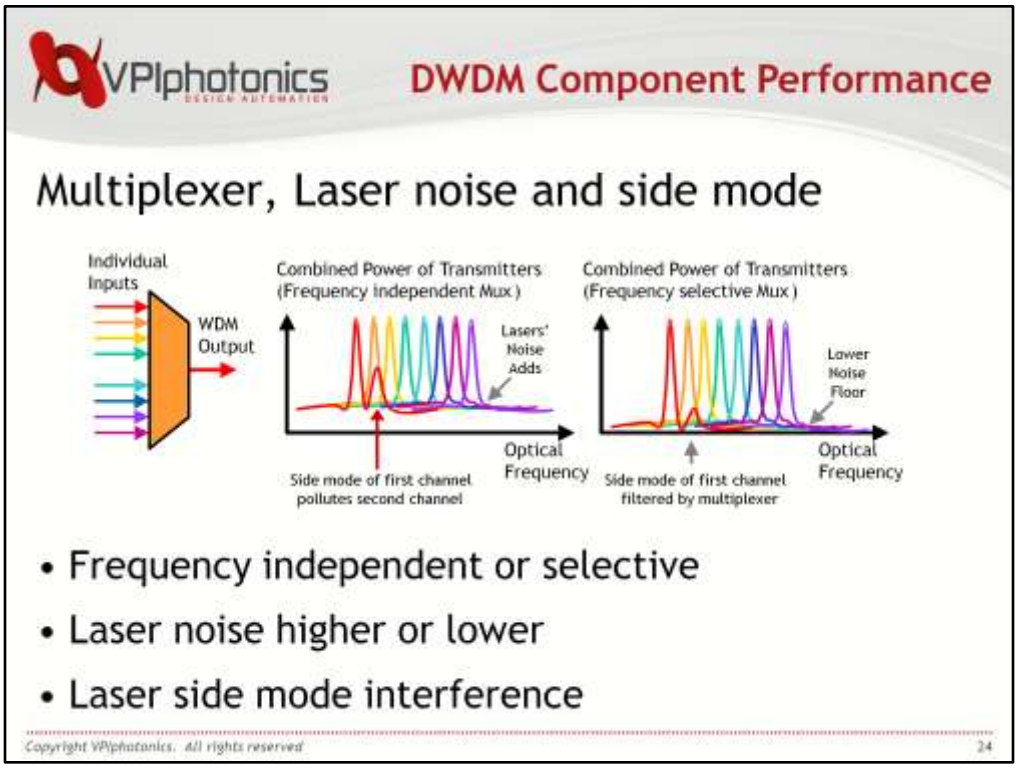




A multiplexer is a device that takes in data signals (at different wavelengths) and combines them into one composite signal that is transmitted through a single fiber.

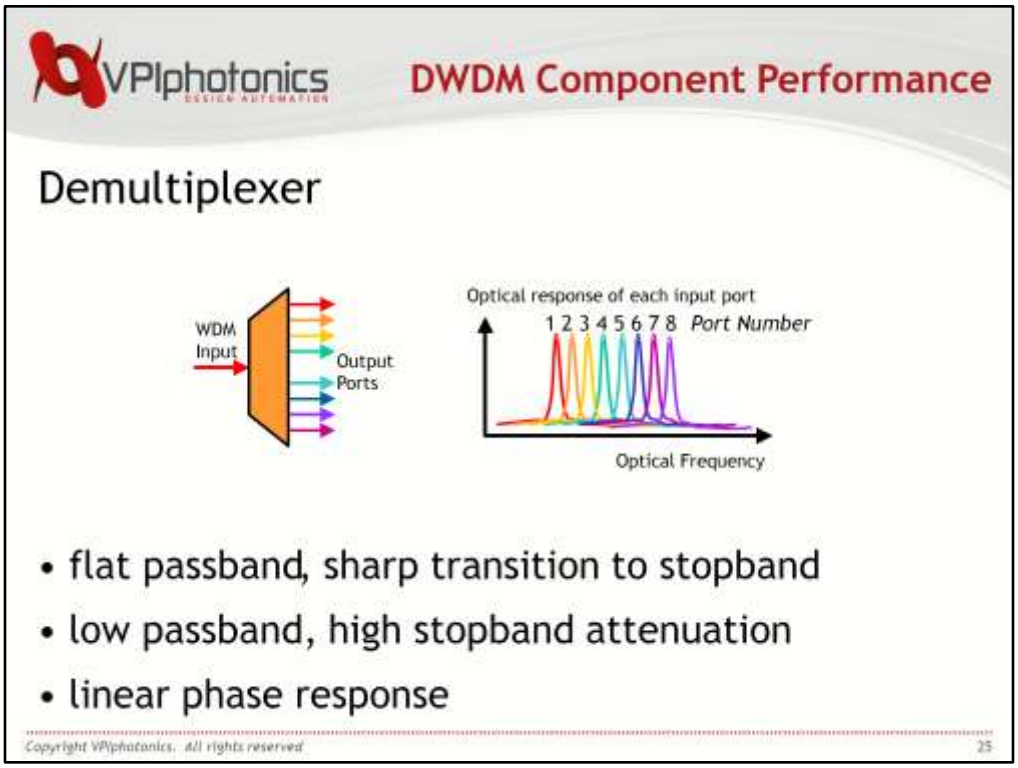
The critical performance parameters of a multiplexer are:
• insertion loss for each channel (the lower the better)
• rejection of adjacent channels (the higher the rejection, the lower the interference)
• passband width of each channel

There are two types of multiplexers – frequency selective and frequency independent (see next slide).



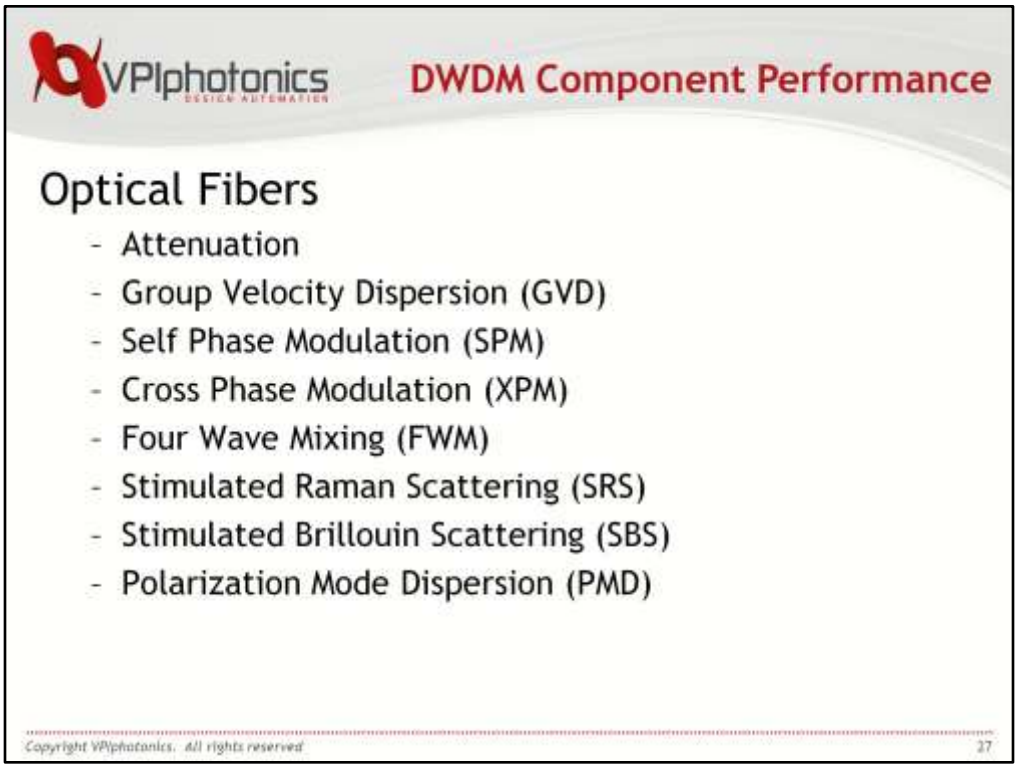
Frequency selective multiplexers filter the individual signals before combining them, whereas the frequency independent multiplexers simply combine all the signals. There are advantages and disadvantages to both types (challenge the students to name these, and state why).

This slide shows how the performance of the WDM system can be affected by the performance of the laser and the multiplexer.

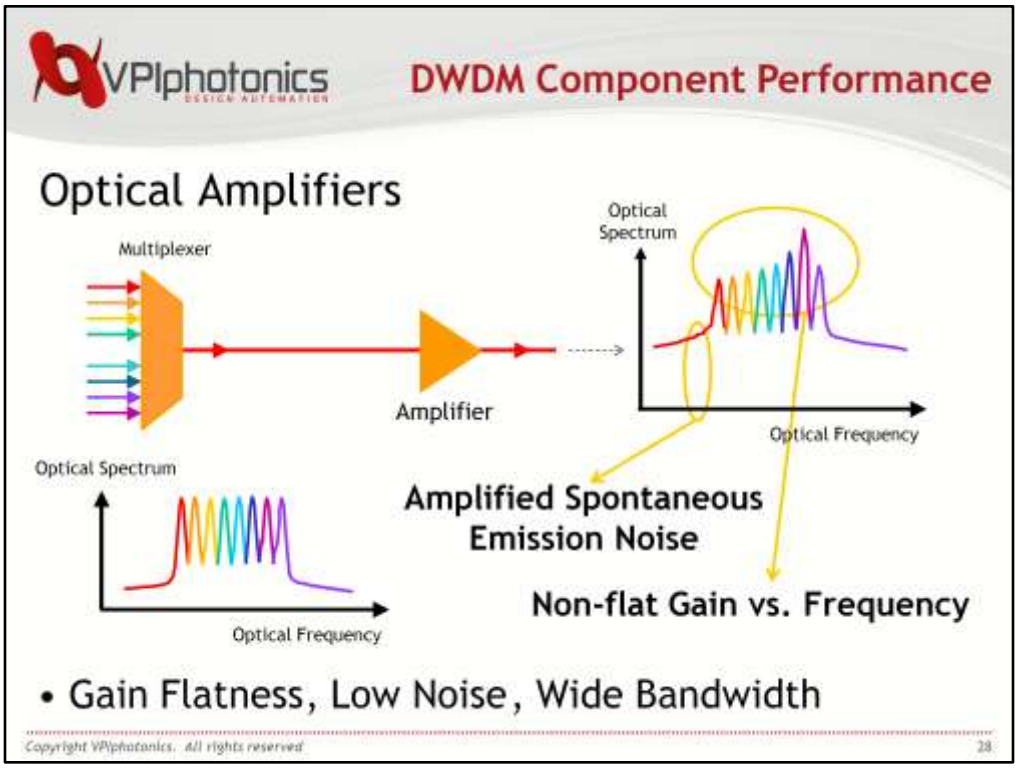


A demultiplexer does the reverse of a multiplexer – it takes a composite signal and separates the different data signals back to their individual wavelength channels.

The critical performance parameters of a multiplexer are:
•Flat passband
•Sharp transition from passband to stopband (which enables the closer spacing of adjacent channels)
•Low passband loss, and high stopband loss (for obvious reasons)
•Linear phase response (to reduce dispersion caused by the demultiplexer, which can accumulate if the signals pass through many demultiplexers in the case that each has non-linear phase response)



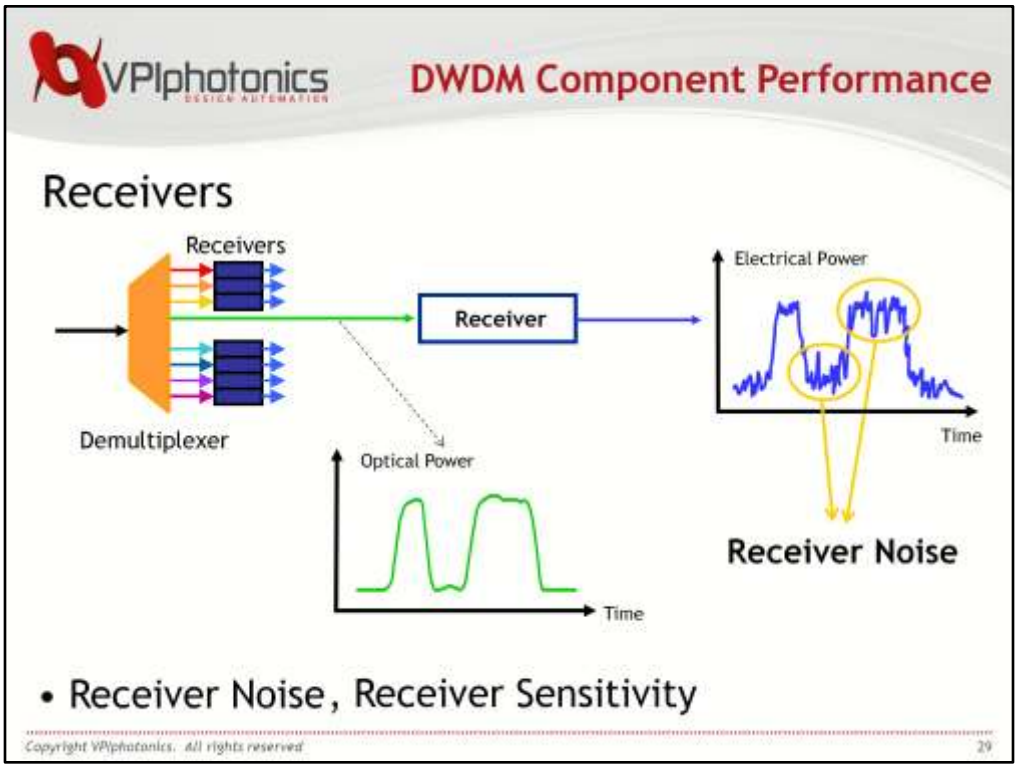
Here is a list of factors that determine the performance of optical fibers in WDM applications. The detailed discussion of the various items on the list is done in separate modules dedicated to optical fibers.



Optical amplifiers, in practice, do not provide the same uniform gain over a wide bandwidth.

The critical performance parameters of an amplifier are:
•Gain flatness across the amplification bandwidth
•Wide bandwidth (enabling more channels to be amplified and thus transmitted)
•Low noise

Details of optical amplifiers are covered in another module

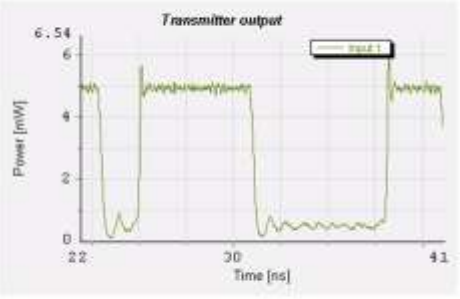


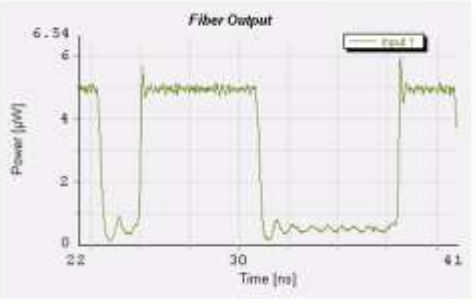
The critical performance parameters of a receiver are:

- Low receiver noise
- High receiver sensitivity (enabling weaker signals to be detected)

Fiber Attenuation

- L=150 km



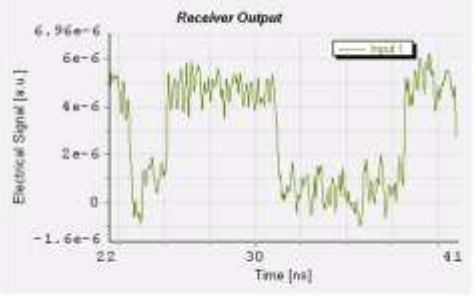


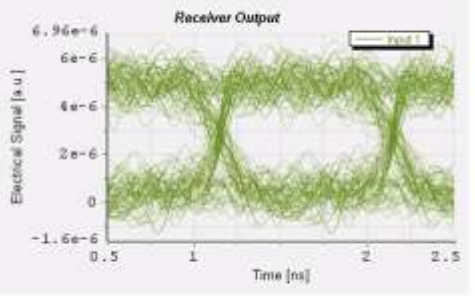
30

Run the setup FOC2_1 as it is.
Three scope traces should appear, and should look like those shown below and the one on the bottom left corner of the next page.

Fiber Attenuation

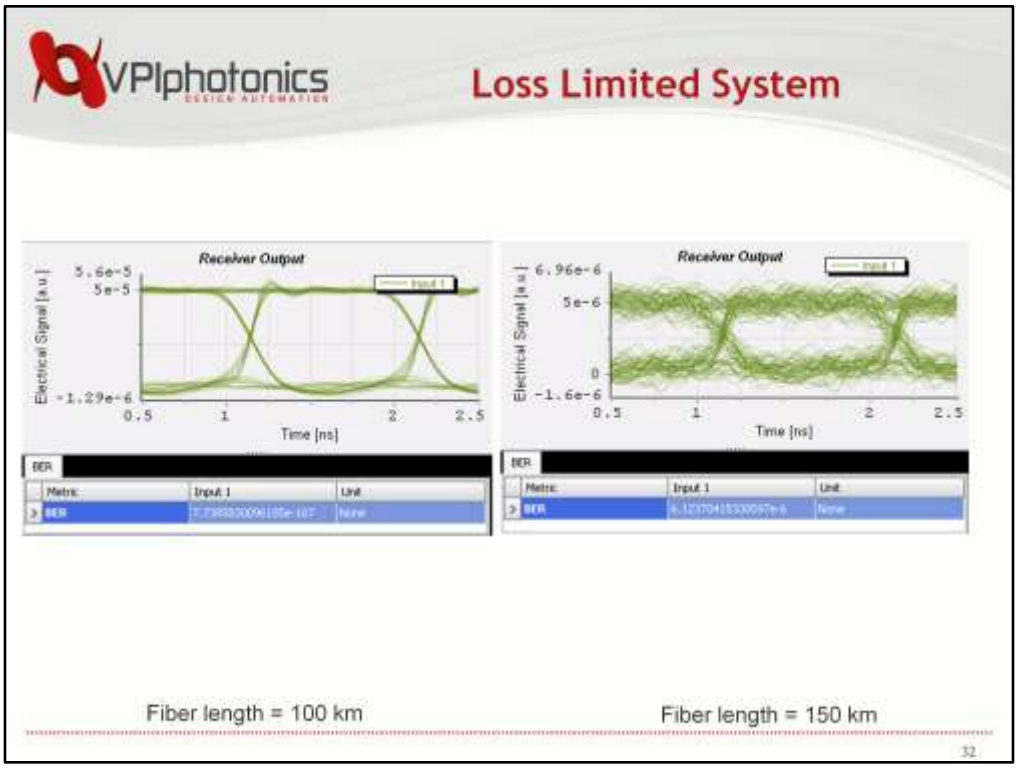
- Receiver output, L=150 km





31

Shown below are the scope trace of the electrical signal detected by the receiver, and the associated eye diagram.

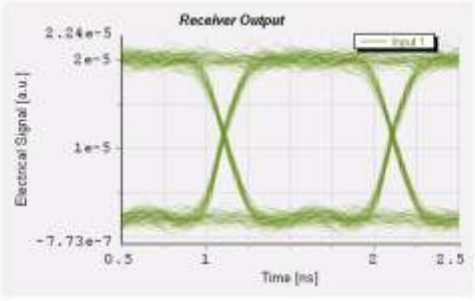


Set the FiberNLS—***Length*** **=** `10.0e3`. Rerun, and obtain the eye diagram of the received electrical signal (shown below in the figure on the left). Use the built-in BER estimator of the scope to find the BER.

Repeat the step above, increasing the fiber length by 10 km each time, until the fiber length is 150 km. Make a plot of the BER vs fiber length (transmission distance).

Loss Limited System: Solution 1

- increasing the output power of the transmitter from 5.0e-3 W to 20.0 e-3 , the fiber length is set to 150 km

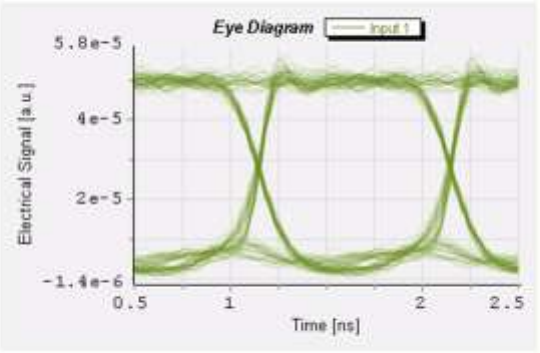


33

To overcome the loss limit, a simple solution is to increase the output power of the transmitter.
Change the parameter *LaserRateEqSM—**PowerNorm*** from `5.0e-3` W to `20.0 e-3` W. Make sure that the fiber length is set to 150 km, and then rerun the simulation. The resultant eye diagram is shown below.

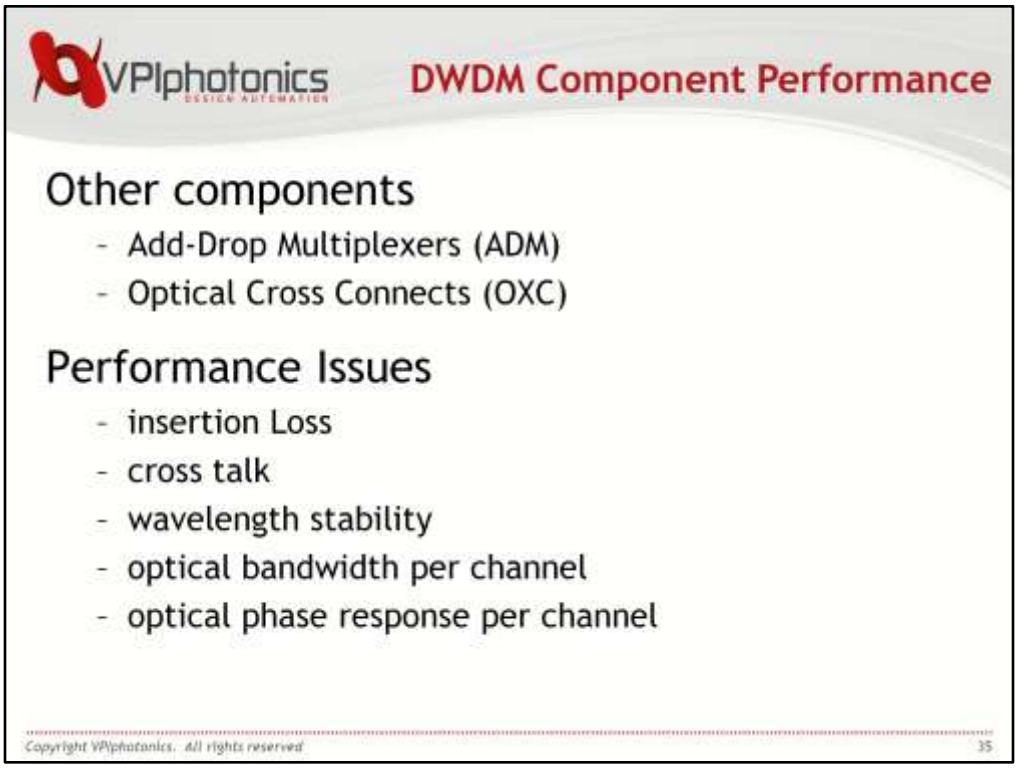
Loss Limited System: Solution 2

- Using optical amplifiers, L=150 km
- .

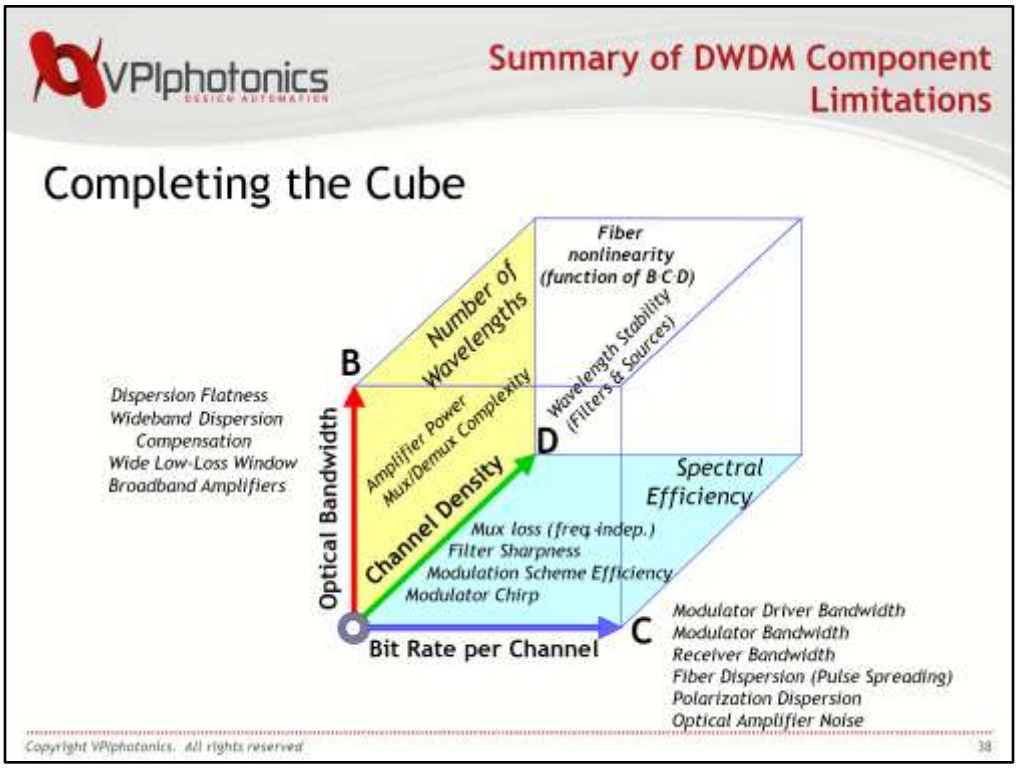


34

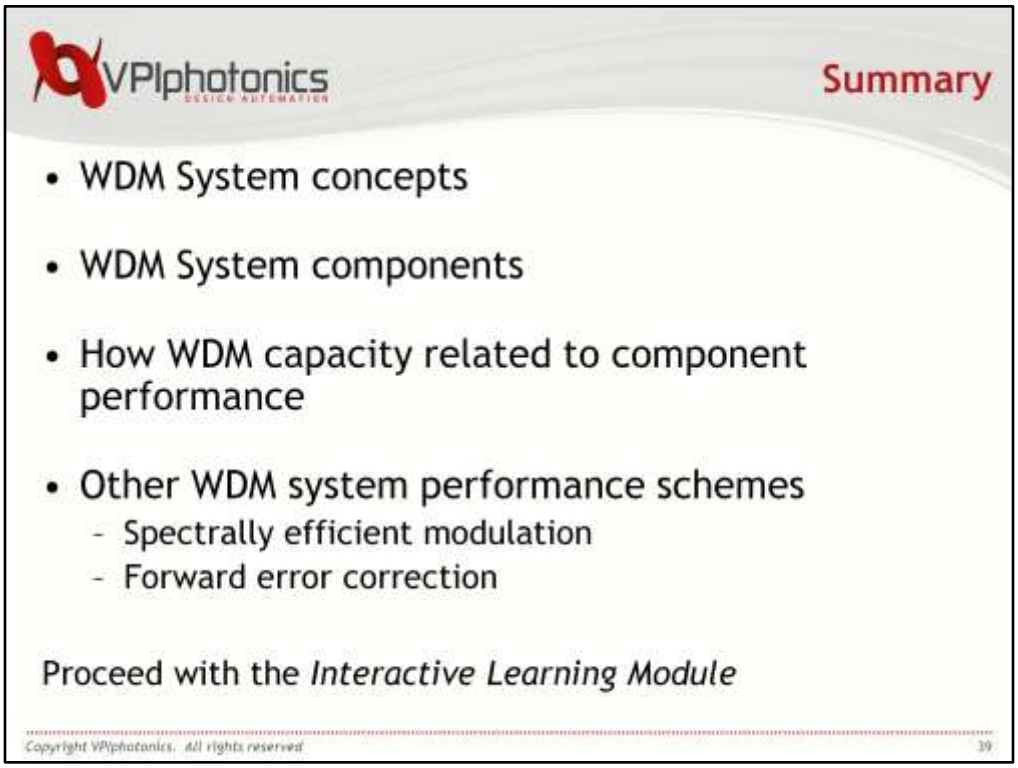
Run FOC2_2. The eye diagram for a signal after 150 km of transmission with fiber attenuation compensated for by an optical amplifier is shown below. There is definite improvement in this case, compared to the case (shown earlier) where no optical amplifier was used.



Here is a list of other components (covered in separate modules), and their performance issues.



The cube picture filled in with component performance factors.



Summary of the key ideas of the lecture part of this module